

6. Write a program to implement Naïve Bayes algorithm in python and to display the results using confusion matrix and accuracy.

CODE:

```
[6]
✓ Os
import matplotlib.pyplot as plt
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.naive_bayes import GaussianNB
from sklearn.metrics import confusion_matrix, accuracy_score, ConfusionMatrixDisplay
# Load Iris dataset
iris = load_iris()
X = iris.data
y = iris.target
# Split dataset
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=1
)
# Train Naive Bayes model
model = GaussianNB()
model.fit(X_train, y_train)
# Predict
y_pred = model.predict(X_test)
# Confusion Matrix
cm = confusion_matrix(y_test, y_pred)
# Display Confusion Matrix as Image
disp = ConfusionMatrixDisplay(confusion_matrix=cm,
                              display_labels=iris.target_names)

disp.plot(cmap="Blues")
plt.title("Confusion Matrix")
plt.show()
# Accuracy
accuracy = accuracy_score(y_test, y_pred)
print("Predicted Values:")
print(y_pred)
print("\nActual Values:")
print(y_test)
print("\nAccuracy:", accuracy * 100)
```

OUTPUT:

